



Mani Rani
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Integrated Circuits & Systems

24D0532
Ph.D.
Gender: Female

Examination	University	Institute	Year	CPI / %
Ph.D.	IIT Bombay	IIT Bombay	2028 (Expected)	9.34
M.Tech	IIT Jodhpur	IIT Jodhpur	2022	7.78
B.Tech	Central University	SLIET, Punjab	2018	8.52

EXPERIENCE (AUG'22 – AUG'24)

- **Chipspirit Technologies** | *Bengaluru* www.chipsprit.com
 - **Worked on Encryption and Decryption Based Project**
 - * Generated the bitstream for BD and RTL Based Projects and did the hardware testing on **ku11p FPGA** board.
 - * Debugging on hardware by adding the **ILA** and **VIO** signals.
 - * Checked the throughput and losses in the data transfer by using the **iperf3** commands.
 - * Written the **RTL**, simulated and synthesized using **Vivado**.

AREAS OF INTEREST

- ASIC Design • Physical Design • Digital VLSI Design • RTL Design & Verification • FPGA

SCHOLASTIC ACHIEVEMENTS

- Selected as a member of the **8-member ARC SoC-B team from IIT Bombay** for the prestigious **1-TOPS (1 TapeOut Per Student)** program organized by the **VLSI Society of India (VSI)**, **ChipIN Centre**, and **C-DAC**; one of only **37 teams** selected nationwide from over **550 proposals** and **3,739 students**.
- Secured **AA** grade in **HDL (Hardware Description Language)** course.

MAJOR PROJECT AND SEMINAR (Ph.D.)

- **Design of Acquisition Module for the NavIC Receiver Chip** *(June'25 – present)*
Guide: Prof. Sachin B. Patkar, Electrical Engineering, IIT Bombay
- **Objective:**
 - **ASIC** design and implementation of the acquisition block using Cadence tools for **GPS/NavIC** receivers, optimized for **performance and power efficiency**.
- **Completed Work:**
 - Designed an **FSM-based serial search GPS acquisition architecture** in **Verilog RTL**, targeting **ASIC** implementation for navigation receivers such as **GPS/NavIC**, covering PRN code phases and Doppler bins.
 - Implemented the complete acquisition datapath, including **PRN generation (G1/G2 LFSRs)**, code-phase rotation, multi-bin Doppler search, and **MAC-based correlation**.
 - Developed a **fixed-point carrier NCO** using phase accumulators and **0.5° resolution sine/cosine LUTs**, implemented in Verilog RTL for carrier generation in the GPS acquisition module.
 - Verified the design through **RTL simulation** and **FPGA prototyping** on **PYNQ-Z2** using real input data.
 - Completed **ASIC implementation** using Cadence **Genus** and **Innovus**, achieving **timing closure**, **zero DRC violations**, and **3.05 mW** total post-layout power, along with comprehensive area, timing, and power analysis.

KEY ACADEMIC PROJECTS AND ASSIGNMENTS

- **AXI4-Lite Signal Generation Module for interfacing with AXI4-Lite BRAM** | *Course Project (Feb'25)*
Course: VLSI Design, Prof. Laxmeesha Somappa
 - Designed an **AXI4-Lite master** testbench interfacing with **AXI-BRAM** for **read/write** operations.
 - Designed a **Verilog** module that generates **AXI4-Lite** signals for **read/write** operations for the **AXI-BRAM** interface and integrated a **16-bit Dadda multiplier** within the module for testing.
 - Validated the design via simulation and on **PYNQ-Z2 FPGA**, capturing **AXI** transactions and multiplier output.
- **Standard Cell Characterization using the Sky130 Open-Source PDK** | *Course Project (Sep'24)*
Course: VLSI Design, Prof. Laxmeesha Somappa
 - Designed **DFF**, **NOR2**, and **Buffer** standard cells in **Sky130 PDK**, ensuring **DRC/LVS** compliance.
 - Used **Ngspice** to extract input **capacitances** and **static power** for various input combinations.
 - Characterized **dynamic power** and cell delays (**fall/rise**) for load capacitances of 10 nF, 100 nF, and 1000 nF at input slew rates of 0.5 ps, 10 ps, and 100 ps.

- **Fault Detection in 74181 ALU Using SAT-Based Testing** | *Course Assignment* (Nov'24)
Course: *Testing and Verification of VLSI Circuits, Prof. Madhav P Desai*
 - Developed a **SAT**-based testing approach to detect **stuck-at** faults in the **74181 ALU**.
 - Generated **CNF** representations using a custom **Python** program to identify **test patterns** covering faults.
- **Design and Implementation of a Dual-Clock Asynchronous FIFO** | *Self Project* (Feb'25)
 - Designed an **Asynchronous FIFO** for efficient data transfer between two distinct **clock domains** in **Verilog**.
 - Realized **binary-to-Gray encoded read/write pointers** for **CDC-safe synchronization**, minimizing multi-bit transitions, mitigating **metastability**, and validating correct FIFO operation through simulation.
- **Design of Decode and ALU Modules for RISC-V Architecture** | *Course Project* (Apr'25)
Course: *VLSI Design Lab, Prof. Laxmeesha Somappa*
 - Implemented **Decode** and **ALU** blocks of a **RISC-V** core in **Verilog**, supporting full **RV32I** instruction decoding using **Xilinx Vivado**.
 - Verified the design on **PYNQ-Z2 FPGA** for functional correctness and timing validation.
 - Completed **ASIC implementation** in **OpenLane** for the decode and ALU modules with no violations.

RELEVANT COURSES TAKEN

- VLSI Design and VLSI Design Lab
- VLSI CAD
- Hardware Description Language
- Testing and Verification
- Introduction to Machine Learning
- Embedded Systems

CERTIFICATIONS

- **VLSI SoC Design using Verilog HDL** *Duration – 4 weeks at Maven Silicon*
- **Deep Learning Based Hardware Systems Design** *Duration – 4 weeks at LNM Institute*
- **Scientific Training in Ubiquitous Division at CSIR-CSIO Chandigarh, India** *Duration – 6 weeks*

TECHNICAL PROFICIENCY

- **Programming:** Verilog, VHDL.
- **EDA & Simulation Tools:** Cadence Genus, Cadence Innovus, Cadence Virtuoso, OpenLane, Magic Layout, Yosys, Netgen, GTKWave, Icarus Verilog, Ngspice, LTSpice, MATLAB.
- **FPGA & Platforms:** Xilinx Vivado, Quartus Prime Lite, PYNQ-Z2, Docker.

CONFERENCES & WORKSHOPS

- Attended the **39th International Conference on VLSI Design (VLSID 2026)** and **25th International Conference on Embedded Systems**, Pune, India, **3–7 January 2026**.

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant: Maker Space Lab** (Aug'24 – Nov'24)
 - Assisted in conducting laboratory sessions and evaluating students for practical experiments.
- **Teaching Assistant: Digital Systems** (Jul'25 – Nov'25)
Course Instructor: *Prof. Debanjan Bhowmik, Electrical Engineering, IIT Bombay*
 - Assisted students with coursework, doubt-clearing sessions, and evaluation activities.
- **Teaching Assistant: Microprocessors** (Jan'26 – May'26)
Course Instructor: *Prof. Shalabh Gupta, Electrical Engineering, IIT Bombay*
 - Assisted students in laboratory sessions, assignments, examinations, and doubt-clearing sessions.
- **Technical Councillor, Hostel 10, IIT Bombay** (Jun'26 – Present)
 - Managed the **Hostel 10 website**, **LinkedIn page**, and **GitHub repository**; coordinated technical activities and promoted student participation in technology initiatives.
- **Hostel Sports Secretary, IIT Bombay** (June'25 – June'26)
 - Organized inter-hostel sports events, coordinated teams, and promoted student participation in sports activities.
- **Advisory Body Member, Happy Club, SLIET Longowal** (2015 – 2018)
 - Supported social service initiatives including **PaathShala**, **Koshish**, and **Wall of Kindness** through volunteer coordination and community outreach.

EXTRACURRICULARS & HOBBIES

- **Hobbies:** Table Tennis | Social Work (Volunteering) | Traveling.